QUAD oscillators LTC6902IMS irradiated failed to pass the post irradiation tests, therefore the device needs to be removed from the design. The rest of the components (DAN217T146CT Schottky diodes, 540BAA100M000BBG oscillators and INA333AIDRG instrumentation amplifiers) passed the post radiation tests with no issues.

5 Conclusions

Radiation test campaigns took place to qualify each of the components of the DB6 design to fulfill the HL-LHC requirements. The outcome of the tests have proved that the design is TID and SEL-resistant. The extra protection added to the design for SEL appearances in the form of a over-current circuit has to be removed due to the shift in the threshold of the MOSFETS used in the implementation. Similarly, the QUAD oscillators used to de-couple the phase of the DC-DC converters have to be removed from the design, and extra studies has to be performed to assess the impact of this design modification on the board noise. The removal of the QUAD oscillators and the over-current circuit will make the design sufficiently radiation-hard for the expected NIEL fluence of the full life of the HL-HLC. Future plans include producing extra prototypes that include the aforementioned modifications, performing Single Event Upset tests, and following-up in the tests to finalize the selection of the SFP+ variant to be used with the final board design. After the design is modified accordingly to the test results, prototypes will be produced and tested as part of the road map for the production phase of the project. A total of 930 DB6 will be produced as the contribution of Stockholm University to the HL-LHC era for TileCal.

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